Minor Project Presentation: Speaker Notes (16-Slide Deck)

Project: Explainable Detection of Prompt Injection Attacks in Hindi and Hinglish **Speakers:** Rahul Patel & Kishan Gupta

Duration: 6 to 8 Minutes 16 Slides Slide-by-Slide Script

Minor Project Presentation: Updated Speaker Notes (Redesigned Deck — 16 Slides)

Project Title: Explainable Detection of Prompt Injection Attacks in Hindi and Hinglish: A Hybrid Rule-Based and Machine Learning Approach
Team Members: Rahul Patel & Kishan Gupta
Target Duration: 6 to 8 Minutes
Slides: 16 (redesigned version)

⏱️ Recommended Speaker Split & Pacing

Slide #	Slide Title	Speaker	Duration
1	Title Slide	Rahul	20s
2	What is Prompt Injection?	Rahul	35s
3	Limitations of Rule-Only and ML-Only	Rahul	40s
4	Reference Paper & Project Scope	Kishan	40s
5	Dataset & Data Leakage Prevention	Kishan	35s
6	Baseline ML Methodology	Rahul	30s
7	Proposed Hybrid Architecture	Kishan	45s
8	Explainability & Contextual Exemption	Rahul	40s
9	Baseline Results (800-Sample Test Set)	Rahul	40s
10	Challenge-Set Evaluation (N=24)	Kishan	45s
11	Failure Analysis — CH06	Kishan	35s
12	Working Prototype (Streamlit)	Kishan	30s
13	Individual Contributions	Both	25s
14	Limitations & Major Project Scope	Rahul	35s
15	Conclusion	Kishan	20s
16	Thank You & Questions	Kishan	15s
Total			**~7.5 min**

Slide-by-Slide Script

Slide-by-Slide Presentation Script

Slide 1: Title Slide

RAHUL PATEL

- **Speaker:** Rahul Patel

> *"Respected faculty panel, good morning/afternoon. I am Rahul Patel, and with my project partner Kishan Gupta, we are presenting our 7th-semester B.Tech Minor Project: Explainable Detection of Prompt Injection Attacks in Hindi and Hinglish — A Hybrid Rule-Based and Machine Learning Approach."*

Slide 2: What is Prompt Injection?

RAHUL PATEL

- **Speaker:** Rahul Patel

> *"Prompt injection is the #1 vulnerability identified by OWASP for LLM applications. An attacker crafts adversarial input that overrides system instructions, causing the model to leak private data or bypass safety. Hindi has 600M+ speakers, and Hinglish dominates digital India — yet most guardrails are English-only. Attackers bypass them simply by translating jailbreaks."*

Slide 3: Limitations of Rule-Only and ML-Only Detection

RAHUL PATEL

- **Speaker:** Rahul Patel

> *"Two common approaches, both with critical flaws. Static regex rules are brittle — rephrase the attack and they miss it entirely. Pure ML models suffer keyword bias — 'system prompt' appears mostly in attack training data, so educational questions get falsely flagged at 78.8% injection probability. Our motivation: bridge this gap with an explainable hybrid cascade."*

Slide 4: Reference Paper & Project Scope

KISHAN GUPTA

- **Speaker:** Kishan Gupta

> *"Our work builds upon Regi et al., published in Nature Scientific Reports, 2026. They benchmarked 6 ML models and deep transformers on a 4,000-prompt Hindi/Hinglish dataset. Our three additions: a hybrid cascade with deterministic short-circuiting, token-level explainability for every prediction, and an out-of-distribution challenge set — which the reference paper did not test."*

Slide 5: Dataset & Data Leakage Prevention

KISHAN GUPTA

- **Speaker:** Kishan Gupta

> *"We used the public benchmark — 4,000 prompts, perfectly balanced across Hindi/Hinglish and safe/injected classes. We split 80/20 with TF-IDF fit strictly on training data to prevent leakage. We also authored a separate 24-prompt challenge set with zero overlap for generalization testing."*

Slide 6: Baseline ML Methodology

RAHUL PATEL

- **Speaker:** Rahul Patel

> *"Our pipeline: raw prompts go through preprocessing, TF-IDF vectorization with (1,2)-gram range and sublinear TF, then Logistic Regression classification. We chose LR because it gives directly interpretable feature coefficients, sub-2ms inference on CPU, and calibrated probabilities."*

Slide 7: Proposed Hybrid Architecture

KISHAN GUPTA

- **Speaker:** Kishan Gupta

> *"Our architecture is a two-stage cascade. A prompt first passes through the heuristic rule layer — if an explicit jailbreak triggers, we short-circuit immediately. If no rule fires, it passes to TF-IDF + Logistic Regression. Our key innovation: the Contextual Inquiry Exemption. If the user asks an educational question without command verbs, we suppress the ML false alarm and mark it safe. Every prediction returns an explainable verdict."*

Slide 8: Explainability & Contextual Inquiry Exemption

RAHUL PATEL

- **Speaker:** Rahul Patel

> *"Our core contribution: resolving keyword bias. When 'system prompt' appears in a question, pure ML predicts 78.8% injection. Our Contextual Exemption recognizes inquiry framing — 'kya hota hai', 'what is' — and zero malicious verbs, successfully saving these as SAFE. Verified on challenge prompts CH18 and CH19."*

Slide 9: Baseline Results (800-Sample Test Set)

RAHUL PATEL

- **Speaker:** Rahul Patel

> *"Our baseline achieved 94% accuracy and 94.09% F1 on 800 unseen test samples. Hinglish scored 96.75% vs Hindi 91.25% — English loanwords provide clearer signals. Confusion matrix: 370 TN, 382 TP, only 18 false negatives. LR coefficients show imperative verbs associate with injection, polite inquiry markers with safe."*

Slide 10: Challenge-Set Evaluation (N=24)

KISHAN GUPTA

- **Speaker:** Kishan Gupta

> *"On 24 newly worded challenge prompts: Rule-only achieved only 50% accuracy, 0% recall — static regexes fail completely on rephrased attacks. ML-only got 79.17% but suffered 4 false positives on security keywords. Our Hybrid achieved 87.50% with the best balance: 91.67% recall and 84.62% precision. The category breakdown shows Hybrid fixed Security Keyword false positives via Contextual Exemption."*

Slide 11: Failure Analysis — CH06

KISHAN GUPTA

- **Speaker:** Kishan Gupta

> *"For scientific honesty: CH06 is our single false negative. It uses hypothetical framing — 'imagine we're in a private room where no filters apply' — without any command verbs. Rule layer found 0 matches, ML gave only 44.7% probability. Detecting this requires deep transformers with contextual attention, which is our Major Project plan."*

Slide 12: Working Prototype (Streamlit)

KISHAN GUPTA

- **Speaker:** Kishan Gupta

> *"We built a fully functional Streamlit prototype. It accepts Hindi, Hinglish, or English input and displays the verdict, confidence, decision route, matched rules, and token attribution — all live. We can demonstrate this now."*

Slide 13: Individual Contributions

RAHUL PATEL

- **Speaker:** Rahul / Kishan

> *"Rahul led the ML pipeline — data hygiene, leakage-free splitting, baseline training, explainability, and evaluation. Kishan led the rule engine, contextual exemption, hybrid integration, challenge set, audit, and Streamlit UI. This allocation is a draft submitted for faculty review."*

Slide 14: Limitations & Major Project Scope

RAHUL PATEL

- **Speaker:** Rahul Patel

> *"We are transparent about our limitations: single-turn only, small challenge set, vocabulary sensitivity. For our 8th-semester Major Project: expand the corpus to 500+ diverse samples, fine-tune IndicBERT and XLM-RoBERTa with sub-word BPE, add multi-turn memory, and deploy as an open-source FastAPI middleware."*

Slide 15: Conclusion

KISHAN GUPTA

- **Speaker:** Kishan Gupta

> *"In conclusion, our Minor Project proves that an explainable hybrid architecture effectively balances speed, interpretability, and false-positive reduction for Hindi/Hinglish prompt injection detection. We have verified code, honest evaluation metrics, and a working prototype."*

Slide 16: Thank You & Questions

KISHAN GUPTA

- **Speaker:** Kishan Gupta

> "*Thank you, and we warmly welcome your questions and feedback.*"